

OpenStax Calculus Volume 3

Section 4.7 Exercises: Maxima/Minima Problems

Source: <https://openstax.org/books/calculus-volume-3/pages/4-7-maxima-minima-problems>.
Exercises reproduced from OpenStax, licensed under CC BY 4.0.

This document was converted programmatically from OpenStax content; formulas were translated automatically from MathML and may contain conversion errors — verify against the source before relying on them.

For the following exercises, find all critical points.

1. $f(x, y) = 1 + x^2 + y^2$
2. $f(x, y) = (3x - 2)^2 + (y - 4)^2$
3. $f(x, y) = x^4 + y^4 - 16xy$
4. $f(x, y) = 15x^3 - 3xy + 15y^3$

For the following exercises, find the critical points of the function by using algebraic techniques (completing the square) or by examining the form of the equation. Verify your results using the partial derivatives test.

5. $f(x, y) = \sqrt{x^2 + y^2 + 1}$
6. $f(x, y) = -x^2 - 5y^2 + 8x - 10y - 13$
7. $f(x, y) = x^2 + y^2 + 2x - 6y + 6$
8. $f(x, y) = \sqrt{x^2 + y^2} + 1$

For the following exercises, use the second derivative test to classify any critical points and determine whether each critical point is a maximum, minimum, saddle point, or none of these.

9. $f(x, y) = -x^3 + 4xy - 2y^2 + 1$
10. $f(x, y) = x^2y^2$
11. $f(x, y) = x^2 - 6x + y^2 + 4y - 8$
12. $f(x, y) = 2xy + 3x + 4y$
13. $f(x, y) = 8xy(x + y) + 7$
14. $f(x, y) = x^2 + 4xy + y^2$
15. $f(x, y) = x^3 + y^3 - 300x - 75y - 3$
16. $f(x, y) = 9 - x^4y^4$
17. $f(x, y) = 7x^2y + 9xy^2$

18. $f(x, y) = 3x^2 - 2xy + y^2 - 8y$
19. $f(x, y) = 3x^2 + 2xy + y^2$
20. $f(x, y) = y^2 + xy + 3y + 2x + 3$
21. $f(x, y) = x^2 + xy + y^2 - 3x$
22. $f(x, y) = x^2 + 2y^2 - x^2y$
23. $f(x, y) = x^2 + y - e^y$
24. $f(x, y) = e^{-(x^2+y^2+2x)}$
25. $f(x, y) = x^2 + xy + y^2 - x - y + 1$
26. $f(x, y) = x^2 + 10xy + y^2$
27. $f(x, y) = -x^2 - 5y^2 + 10x - 30y - 62$
28. $f(x, y) = 120x + 120y - xy - x^2 - y^2$
29. $f(x, y) = 2x^2 + 2xy + y^2 + 2x - 3$
30. $f(x, y) = x^2 + x - 3xy + y^3 - 5$
31. $f(x, y) = 2xye^{-x^2-y^2}$

For the following exercises, determine the extreme values and the saddle points. Use a CAS to graph the function.

32. [T] $f(x, y) = ye^x - e^y$
33. [T] $f(x, y) = x \sin(y)$
34. [T] $f(x, y) = \sin(x) \sin(y)$, $x \in (0, 2\pi)$, $y \in (0, 2\pi)$

Find the absolute extrema of the given function on the indicated closed and bounded set R .

35. $f(x, y) = xy - x - 3y$; R is the triangular region with vertices $(0, 0)$, $(0, 4)$, and $(5, 0)$.
36. Find the absolute maximum and minimum values of $f(x, y) = x^2 + y^2 - 2y + 1$ on the region $R = \{(x, y) | x^2 + y^2 \leq 4\}$.
37. $f(x, y) = x^3 - 3xy - y^3$ on $R = \{(x, y) : -2 \leq x \leq 2, -2 \leq y \leq 2\}$
38. $f(x, y) = \frac{-2y}{x^2+y^2+1}$ on $R = \{(x, y) : x^2 + y^2 \leq 4\}$
39. Find three positive numbers the sum of which is 27, such that the sum of their squares is as small as possible.
40. Find the points on the surface $x^2 - yz = 5$ that are closest to the origin.
41. Find the maximum volume of a rectangular box with three faces in the coordinate planes and a vertex in the first octant on the plane $x + y + z = 1$.
42. The sum of the length and the girth (perimeter of a cross-section) of a package carried by a delivery service cannot exceed 108 in. Find the dimensions of the rectangular package of largest volume that can be sent.

43. A cardboard box without a lid is to be made with a volume of 4 ft^3 . Find the dimensions of the box that requires the least amount of cardboard.
44. Find the point on the surface $f(x, y) = x^2 + y^2 + 10$ nearest the plane $x + 2y - z = 0$. Identify the point on the plane.
45. Find the point in the plane $2x - y + 2z = 16$ that is closest to the origin.
46. A company manufactures two types of athletic shoes: jogging shoes and cross-trainers. The total revenue from x jogging shoes and y cross-trainers is given by $R(x, y) = -5x^2 - 8y^2 - 2xy + 42x + 102y$, where x and y are in thousands of units. Find the values of x and y to maximize the total revenue.
47. A shipping company handles rectangular boxes provided the sum of the length, width, and height of the box does not exceed 96 in. Find the dimensions of the box that meets this condition and has the largest volume.
48. Find the maximum volume of a cylindrical soda can such that the sum of its height and circumference is 120 cm.